Picture Hanging Puzzle

Math Olympiad Club Zurich

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Problem: Picture Hanging Puzzle, A. Spivak, 1997

We have $n \geq 1$ nails fixed to a wall and a sufficiently long rope wrapped around these nails. We consider here what we call *oriented wrapping*, which means a circuit (with a direction) of the rope that may pass over the nail (loop) in a specific orientation (either clockwise or anticlockwise). To help you imagine the situation, consider the two endpoints of the rope carrying a framed picture on which gravity acts. We strongly encourage you to try to recreate the scenario in the real world.

For integers $k, j \in \mathbb{N}$, saying a configuration has k loops around a nail with a clockwise/anticlockwise orientation means exactly that the circuit passes k times **over** that nail with that same respective orientation, where clockwise is left to right and anticlockwise is right to left. Note that the same nail can have k loops in one orientation and j loops in the other orientation. Note that 1 loop with any orientation may not constitute a $360°$ turn.

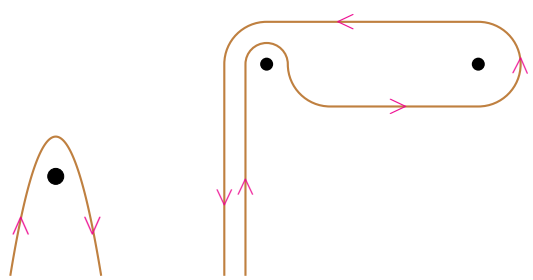

Figure 1: *Examples illustrating that a loop does not need to be a full* $360°$ *turn (left), and a single rope making a clockwise loop and an anticlockwise loop around the same (left) nail, passing via a second (right) nail.*

Consider all oriented wrappings around the n nails satisfying the following two conditions:

1. The circuit is non-trivial; that is, the rope remains securely wrapped (i.e., it does not fall off when all nails are present), or, equivalently, the framed picture does not fall.

2. When **any single nail** is removed (regardless of which one), the entire rope falls from the wall. In practice, some friction might prevent it from falling, but we consider it as "falling" if it is no longer securely wrapped.

The following figures are examples where we have labelled the nails c_0, c_1, and c_2.

Figure 2: *Oriented wrappings for $n = 1$* ***satisfying*** *the properties. The first configuration has a single loop around c_0 (clockwise), while the second has at least two clockwise loops around c_0.*

Figure 3: *Non-trivial oriented wrappings for $n = 2$* ***not satisfying*** *the property 2. The first configuration has 2 loops around c_0 but at least 2 clockwise loops around c_1. The second configuration has at least 2 clockwise loops around both c_0 and c_1.*

Figure 4: *Non-trivial oriented wrappings for $n = 3$* ***not satisfying*** *the property 2. The first configuration has a single clockwise loop around c_0, no loops around c_1, and a single clockwise loop around c_2. The second configuration has at least two clockwise loops around c_0, no loops around c_1, and at least two anticlockwise loops around c_2.*

For any oriented wrapping, we define its *length* to be the number of loops in its circuit. Therefore, the lengths of the oriented wrappings in the above example (Figure 2, 3, 4) are, respectively, for certain $a, b, c, d, e \geq 2$:

$$1,\ a,\ b,\ 2c,\ 2,\ d + e.$$

For each $n \geq 1$, construct an oriented wrapping with the above two properties whose length is bounded by a polynomial in n, more precisely, of length at most $2n^2 - 1$.